

The Relationship between Education and Income across U.S. States

Econ 570 Final Project, Group 32

Rohan Abraham¹ Rohit Sriram² Arjun Bharadhwaj³
University of Wisconsin–Madison

Introduction

Our project looks at whether U.S. states with a higher number of college educated adults also have higher median household incomes, and what the potential implications of this relationship are. We started with a basic correlation, then expanded our analysis to rule out some potential issues with the data. We looked at whether the relationship holds when examining older cohorts who completed their education well before their parents reached their current income levels. We also looked at whether it persists after accounting for differences in cost of living and the types of industries present across states.

We found the initial correlation to be pretty strong. When we ran a basic regression of 2023 state median household income on the share of adults with a bachelor's degree, we got a slope of about \$1,591 per percentage point, with an R^2 of 0.69 (Figure 1). Two additional tests complicated the story, however. First, when we swapped in the college share for adults aged 45 to 64, who finished school 25 to 40 years ago, the slope actually got a little steeper instead of weaker. This is the opposite of what you'd expect if reverse causality were really driving the results. Second, once we controlled for cost of living, the education coefficient dropped by about half. This suggests that a big chunk of the original correlation is just reflecting the fact that high-education states tend to be expensive states too.

Data

We looked at four datasets, all of them are from federal statistical agencies and dated 2023 (or the closest year available). The U.S. Census Bureau's American Community Survey 2019–2023 5-Year Estimates provides Table S1901 (income in the past 12 months) and Table S1501 (educational attainment and median earnings by attainment level) [1, 2]. We also used the Bureau of Economic Analysis's Regional Price Parities for 2023 [3] and its state-by-industry GDP table [4]. From the BEA tables we computed, for each state, a cost-of-living index and the share of state GDP coming from three relatively high-wage sectors: mining, information, and finance and insurance. After dropping Puerto Rico we had 50 states plus DC, for 51 observations. Summary statistics are in Table 1.

¹rvabraham@wisc.edu

²rsriram2@wisc.edu

³bharadhwaj@wisc.edu

Analysis

The cross-state association (Q1)

An OLS regression of state median household income on the bachelor's-or-higher share (Figure 1) outputted a slope of \$1,591 per percentage point and $R^2 = 0.69$. The District of Columbia is far out on the right of the scatter, so we re-ran the regression without it; the slope rose to \$1,964/pp and the R^2 rose to 0.73. Therefore, DC alone is not driving the relationship. Mississippi and West Virginia sat at the bottom of the cloud, with median incomes near \$58,000 and college shares near 25%. Massachusetts and New Jersey sat near the top. A few states sat well above the regression line, including Alaska, Wyoming, and North Dakota. Those states are all energy and mining heavy, which previews the omitted-variable question we look at in Q3.

What can we learn? Threats to identification

A positive cross-state correlation does not by itself identify a causal effect. As we conducted this analysis, we wanted to keep an eye out for two things: reverse causality, where high parental incomes in a state raise children's education in that same state, and omitted-variable bias. We address each one with a separate check.

Reverse causality (Q2): If reverse causality were really driving the relationship, the cross state slope should be steepest for the youngest cohort. This is because their schooling decisions are the most recent and most affected by current parental incomes. We can test this by re-running the Q1 regression with cohort-specific bachelor's shares from S1501. For the 25 to 34 cohort, the slope is \$1,106/pp with $R^2 = 0.56$. For the 45 to 64 cohort, the slope is steeper, \$1,879/pp with $R^2 = 0.74$ (Figure 2). The older cohort finished school 25 to 40 years ago, well before today's parental incomes were earned. Therefore, their education levels are set in advance from the standpoint of today's earnings. The fact that the older-cohort slope is the larger one is the opposite of what reverse causality would produce. This is a robustness check rather than a clean causal identification. However, it does push the remaining worry away from reverse causality and toward standard omitted-variable bias.

Omitted-variable bias (Q3): We added controls one at a time and watched the bachelor's coefficient (Figure 3 and Table 2). Spec (1) is the baseline regression from Q1 at \$1,591/pp. Spec (2) adds the Regional Price Parity, a state-level cost-of-living index. The education coefficient drops by about half to \$807/pp, while R^2 rises from 0.69 to 0.84. A substantial part of the raw correlation is just that high-education states tend to be high-cost states. Spec (3) adds the share of state GDP from mining, information, and finance, and the education coefficient barely moves. Spec (4) swaps in cost-of-living-adjusted real income on the left hand side, and the education coefficient is \$968/pp, still positive and significant but about 40% below the original number. The simple cross-state correlation overstates the per-degree income gain by something like a factor of two once we account for the obvious confounders.

Conclusions and directions for future research

The short answer to our main question is yes: each extra percentage point of bachelor's-or-higher share lines up with roughly \$1,600 of state median household income. The robustness checks make the answer more careful. The older-cohort

regression is at least as steep as the baseline, so reverse causality does not look like the main driver of the cross-state correlation. However, controlling for cost of living cuts the education coefficient roughly in half. Therefore, a lot of the raw correlation is a price-level effect rather than a return to skills.

A few directions for future work: our analysis is at the state level, so the coefficient we estimate is not the individual-level return to schooling. A natural next step would be to use ACS public-use microdata so we can regress on individual workers instead of state averages. A cleaner causal identification would need an instrumental-variables design built on policy changes that affect schooling but are not themselves driven by current parental income. Pulling apart what is inside the cost-of-living and industry controls (state labor market features that drive within-group wage differences) would also be a good follow-up

A Figures and tables

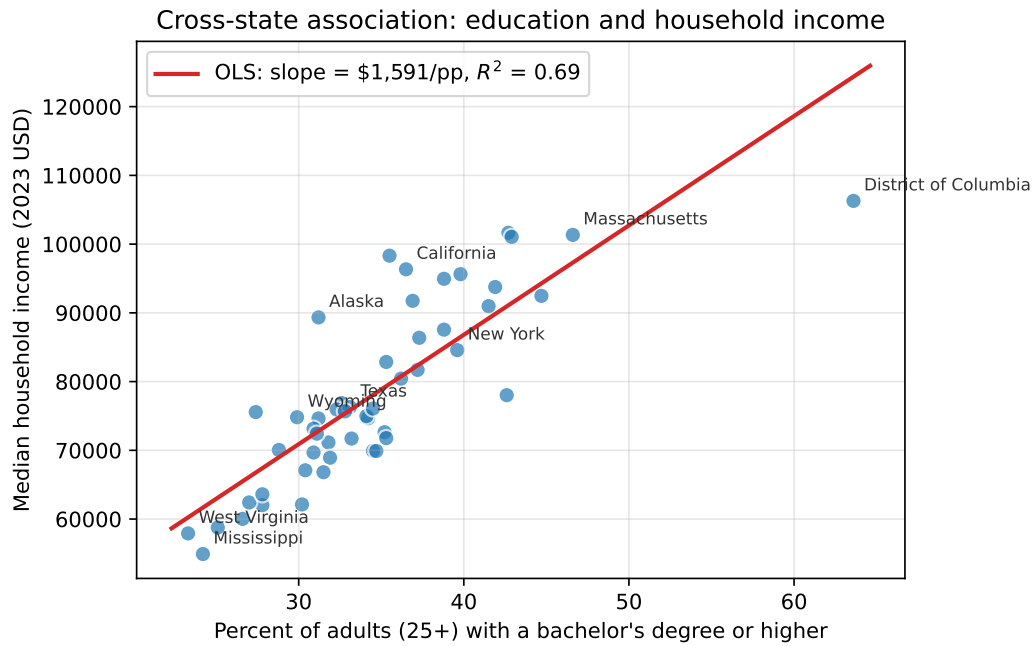


Figure 1: Q1. State median household income vs. percent of adults aged 25+ with a bachelor's degree or higher, ACS 2019–2023 5-Year Estimates. Red line: bivariate OLS over all 51 observations. Selected outliers labeled.

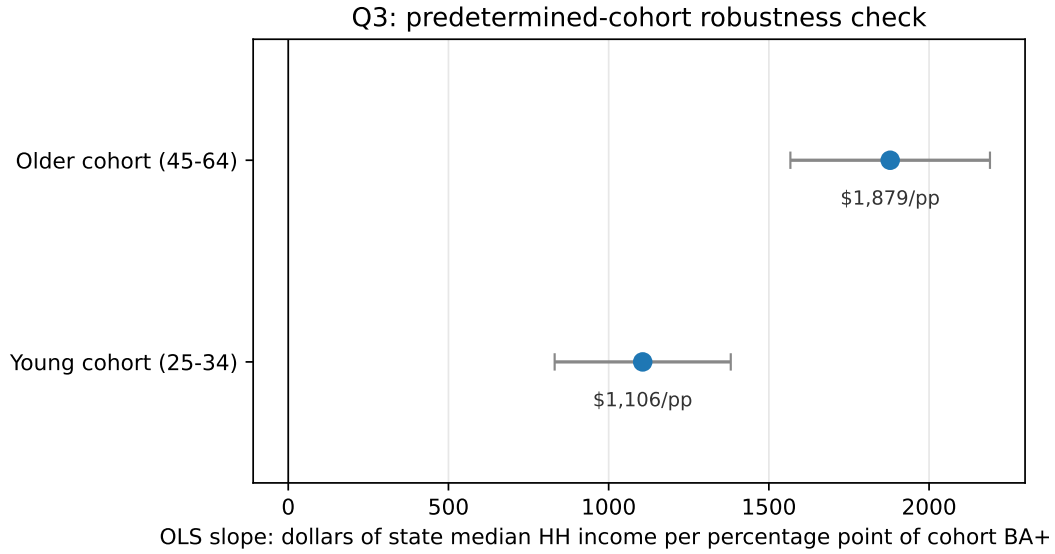


Figure 2: Q2. OLS slope of state median household income on cohort-specific bachelor's-or-higher share, with 95% confidence intervals. The older cohort's education was set decades before today's parental incomes were earned; its slope is larger than the young cohort's, the opposite of what reverse causality would predict.

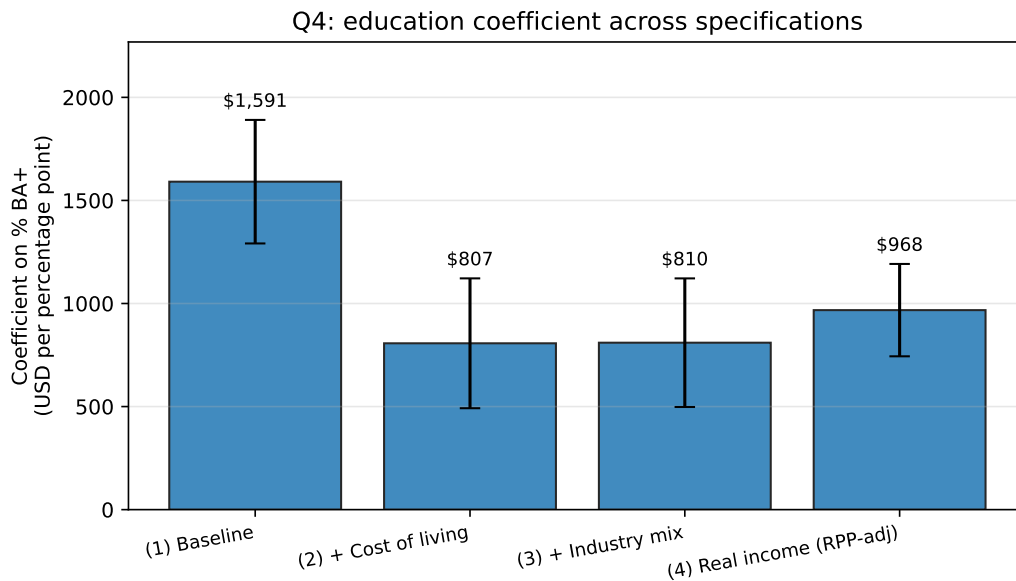


Figure 3: Q3. Coefficient on percent BA+ across four nested specifications, with 95% confidence intervals. Adding the cost-of-living index (Regional Price Parity) roughly halves the coefficient; further adding industry mix barely moves it. Spec (4) uses RPP-adjusted real income on the left-hand side.

Table 1: Summary statistics for the state-level panel ($n = 51$, 50 states + DC).

Variable	N	Mean	SD	Min	Max
Median household income (USD)	51	78,195	12,978	54,915	106,287
Real (RPP-adjusted) HH income (USD)	51	80,225	8,604	63,272	97,180
% adults 25+ with BA+	51	34.59	6.77	23.30	63.60
% adults 25–34 with BA+	51	37.66	8.78	25.40	74.80
% adults 45–64 with BA+	51	33.13	5.94	21.70	53.10
Regional Price Parity (US=100)	51	96.98	7.22	86.79	112.19
% state GDP from finance+info+mining	51	13.12	4.97	6.57	28.29

Table 2: Q3. Sequential OLS regressions; coefficients with standard errors in parentheses. Specifications (1)–(3) use median household income; specification (4) uses RPP-adjusted real income on the left-hand side. All regressions include 51 observations (50 states + DC).

Variable	(1) Baseline	(2) +RPP	(3) +Industry	(4) Real income
% BA+, age 25+	1,591*** (153)	807*** (161)	810*** (159)	968*** (114)
Regional Price Parity		1,014*** (151)	1,001*** (149)	
% GDP from high-wage sectors			209 (150)	259 (156)
R^2	0.69	0.84	0.85	0.61

*** $p < 0.01$.

References

- [1] U.S. Census Bureau. *American Community Survey 2019–2023 5-Year Estimates, Table S1501: Educational Attainment*. Retrieved from <https://data.census.gov/table/ACSST5Y2023.S1501>.
- [2] U.S. Census Bureau. *American Community Survey 2019–2023 5-Year Estimates, Table S1901: Income in the Past 12 Months*. Retrieved from <https://data.census.gov/table/ACSST5Y2023.S1901>.
- [3] U.S. Bureau of Economic Analysis. *Regional Price Parities by State, 2008–2024 (table SARPP)*. Retrieved from <https://apps.bea.gov/regional/zip/SARPP.zip>.
- [4] U.S. Bureau of Economic Analysis. *Gross Domestic Product by State by Industry, 1997–2025 (table SAGDP2)*. Retrieved from <https://apps.bea.gov/regional/zip/SAGDP.zip>.